

# CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of B. Tech. (CL) Examination  
November - 2012

## CL 207 Concrete Technology

Date: 03.11.2012, Saturday

Time: 01:30 p.m. To 4:30 p.m.

Maximum Marks: 70

### Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

### SECTION - I

- Q - 1 (a) "Water is essential ingredient of concrete, it also affect the durability of concrete." – Justify the statement with reason and examples. 03
- (b) State four special concrete of modern era with its ingredients. 04
- Q - 2 (a) Explain the manufacturing process of Portland cement with neat sketch. 04
- (b) Name laboratory tests to measure soundness of cement. Name the chemicals responsible for unsoundness of cement. 03
- (c) State the properties and use of Rapid Hardening Cement. 04
- (d) What is Fly Ash? Explain influence of fly ash in mass concrete with respect to cement. 03

### OR

- Q - 2 (a) What is setting time? Explain the term initial and final setting time of Ordinary Portland Cement. 03
- (b) Describe the role of gypsum in the hydration process of cement. 04
- (c) State the properties and use of Sulphate resisting cement. 04
- (d) From where silica fumes is obtained? Explain its influence on properties of fresh concrete 03
- Q - 3 (a) Explain scope and role of air entraining admixture in brief. 03
- (b) What is difference between additives and admixtures? 04
- (c) Explain alkali aggregate reaction in aggregate. State factors affecting alkali aggregate reaction. 07

### OR

- Q - 3 (a) What are admixtures? Explain the benefits of using admixture in concrete. Also explain water reducing admixture and its effect on properties of fresh concrete. 07
- (b) Explain various mechanical properties of aggregate. 07

## SECTION - II

- Q - 4 (a) In mix proportioning, generally we use minimum quantity of water, why? 03  
(b) Find the total yield of concrete in liter for one bag of cement with the proportion of 1:1.6:2.4. Consider specific gravity 3.2, 2.7, 2.8 and 1 for cement, FA, CA and water respectively. Entrapped air 2%. 04
- Q - 5 (a) In detail, explain manufacturing process of concrete. 07  
(b) Explain Fiber reinforced concrete in terms of ingredients and properties at its fresh state. 03  
(c) State various methods of mix design. Write method of any one. 04
- OR**
- Q - 5 (a) Write a note on workability of concrete including definition, factors affecting it and methods of its measurement. 07  
(b) "This is the era of ready mix concrete." – Justify the statement. 03  
(c) Explain how moisture correction of moist aggregate is considered in water demand in mix design. 04
- Q - 6 (a) State various causes of permeability. Why durability and permeability correlated with each other? 07  
(b) Why concrete is tested on the 28<sup>th</sup> day after casting? 02  
(c) State factors affecting the workability of concrete. 02  
(d) State various precautionary measures during mass concreting. 03
- OR**
- Q - 6 (a) State factors affecting strength of concrete. 07  
(b) How concrete is classified according to its strength as per IS: 456 – 2000? 02  
(c) State parameters which results in to segregation of concrete. 02  
(d) State disadvantages of fiber reinforced concrete. 03

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